SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ANSWER SCRIPT EVALUATION SHEET

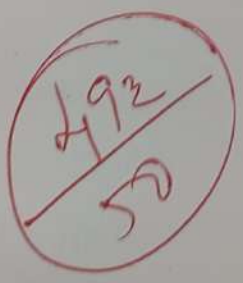
ASSIGNMENT - II

Programme	B.E / B.Tech	**Course Code**	MLA02
Course Name	**Fundamentals of Machine learning.**	**Date**	31/10/2026
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: 196526011 6

TASK 1	Max Marks	Marks Awarded
Task 1	5	
Task 2	2	
Task 3	6	
Task 4	4	
Task 5	3	
Task 6	3	
Task 7	4	
Task 8	4	
Task 9	3	
Task 10	4	
Task 11	4	
Task 12	8	
TOTAL	**50**	

Total in words:



TASK 1 - Design of learning systems :

According to Tom Mitchell, a machine-learning system learns from Experience (E) to perform a task(T) and improve its Performance (P).

Analysis of the five systems:

System	Learning paradigm	Learning task	Experience (E)	performance
a) Loan default prediction	Supervised	classification	10 years of labelled Repayment records	Accuracy, precision and recall.
b) Customer Segmentation	unsupervised	clustering	2 million un-labelled customer record	Silhouette score
c) Credit-card Spending prediction	Supervised	Regression	Historical Spending records	MAE & RMSE
d) ATM Cash-refill policy	Reinforcement	control	Repeated trials and feedback	Maximum reward by balancing cash availability & transport cost.
e) Fraud detection	Semi-supervised	classification	2% labelled & 90% unlabeled transactions	precision, Recall & F1-score

System Most affected by Label Noise :

Loan default prediction (system a) is likely to be most affected by label noise because it depends entirely on labelled historical data. If some loan records are incorrectly labelled as default or non-default, the model may learn incorrect patterns & make wrong predictions.

In fraud detection, the large amount of unlabelled data may help the model discover additional patterns, reducing its dependence on labelled examples.

Conclusion:
Different machine learning paradigms are selected based on the type of data available and the task to be performed.

2. TASK 2 - Bayes' Theorem

Given:
prevalence = 2% = 0.02
Sensitivity = 95% = 0.95
Specificity = 90% = 0.90
False positive rate = 1 - 0.90 = 0.10

Formula : Bayes' theorem

$$P(D/+) = \frac{P(+/D)P(D)}{P(+/D)P(D) + P(+/\bar{D})P(\bar{D})}$$

Where :
- $P(D/+)$ = probability of having disease after a positive test
- $P(+/D)$ = sensitivity
- $P(D)$ = Disease prevalence.

Substitution:

$$P(D/+) = \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.019 + 0.098} = \frac{0.019}{0.117} = 0.1624$$

Answer : $P(D/+) = 16.24\%$

nalysis :

the probability is much lower than the sensitivity because the disease prevalence is only 2%. Even though the test correctly detects 95% of diseased people, a large number of healthy people can also test positive. therefore, a positive test result does not necessarily mean that the person has the disease.

3. TASK 3 - Joint Probability Distribution

Given joint probability distribution:

X/Y	Y=0	Y=1
X=0	0.10	0.20
X=1	0.15	0.25
X=2	0.20	0.10

a) Marginal distribution

The marginal probability of X is obtained by adding the joint probabilities across each row:

$$P(X=x) = \sum_{y} P(X=x, Y=y)$$

$P(X=0) = 0.10 + 0.20 = 0.30$

$P(X=1) = 0.15 + 0.25 = 0.45$

$P(X=2) = 0.20 + 0.10 = 0.30$

Marginal distribution of X:

X	P(X)
0	0.30
1	0.45
2	0.30

Similarly, marginal distribution of y:

$$P(Y=y) = \sum_x P(X=x, Y=y)$$

$P(Y=0) = 0.10+0.15+0.20 = 0.45$

$P(Y=1) = 0.20+0.25+0.10 = 0.55$

y	$P(Y)$
0	0.45
1	0.55

(b) Mean & variance:

Expected value of X: $E[X] = 0(0.3)+1(0.4)+2(0.3) = \underline{1}$

Second moment: $E[X^2] = 0+1(0.4)+4(0.3) = \underline{1.6}$

Variance: $Var(X) = 1.6-1^2 = \underline{0.6}$

Expected value of y: $E[Y] = 0(0.45)+1(0.55) = \underline{0.55}$

(c) Conditional probability

$$P(X=2 / Y=1) = \frac{0.10}{0.55} = 0.1818$$

(d) Covariance & correlation

$E[XY] = 0+0+0+0.25+0+0.20 = 0.45$

$Cov(X,Y) = 0.45-(1)(0.55) = -0.10$

For y: $Var(Y) = 0.55(0.45) = 0.2475$

correlation:

$$\rho = \frac{-0.10}{\sqrt{0.6 \times 0.2475}} = \underline{-0.259}$$

(e) Independence:

check:

$P(0,0) = 0.10$

$P(X=0)\,P(Y=0) = 0.30 \times 0.45 = 0.135$

since

$0.10 \neq 0.135$

X & Y are not independent

Task 4 - Normal distribution & Maximum Likelihood Estimation

Given:

Mean (μ) = 50, $\sigma^2 = 16$

Standard deviation $\sigma = \sqrt{16} = 4$

The standard normal variable is : $\boxed{z = \frac{x-\mu}{\sigma}}$

1. Find $P(X > 58)$

using the z-score formula:

$$z = \frac{x-\mu}{\sigma} = \frac{58-50}{4} = 2$$

Given, $\phi(2) = 0.9772$

$$P(X>58) = 1-\phi(2)$$
$$= 1 - 0.9772 = \underline{0.0228}$$

2. Find $P(46 < X < 54)$

converting into z-scores:

$$z_1 = \frac{46-50}{4} = -1 \qquad z_2 = \frac{54-50}{4} = 1$$

$$P(46<X<54) = P(-1<z<1)$$
$$= 0.8413 - 0.1587$$
$$= \underline{0.6826}$$

b) Given readings : 12, 15, 11, 14, 13

1. Maximum Likelihood Estimate (MLE) of Mean

$$\hat{\mu} = \frac{\Sigma x_i}{n} = \frac{12+15+11+14+13}{5} = 13$$

$\boxed{\hat{\mu} = 13}$

2. Maximum Likelihood Estimate (MLE) of variance

$$\hat{\sigma}^2 = \frac{\Sigma(x_i - \bar{x})^2}{n} = \frac{(12-13)^2 + (15-13)^2 + (11-13)^2 + (14-13)^2 + (13-13)^2}{5} = \frac{1+4+4+1+0}{5} = 2 = \hat{\sigma}^2$$

3. Calculate unbiased variance:

$$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n-1} = \frac{10}{5-1} = \frac{10}{4} = 2.5$$

$\boxed{s^2 = 2.5}$

5. TASK 5 - Information Theory and Entropy

Given: $P = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$

Step 1 - calculate entropy

Entropy measures the average information content of source

Formula: $H(X) = -\sum p(x) \log_2 p(x)$

Substitution-

$$= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right]$$

$$= \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$H(X) = 1.75$ bits

Step 2 - Entropy of uniform source for four equally likely symbols: $P(x) = \frac{1}{4}$

formula: $H_{uniform} = \log_2 4$

$H_{uniform} = 2$ bits

Step 3 - Calculate redundancy

Absolute redundancy

$$R = H_{max} - H(X)$$

$$= 2 - 1.75$$

$R = 0.25$ bits

Relative redundancy:

$$\therefore R_r = 1 - \frac{H(X)}{H_{max}} = 1 - \frac{1.75}{2} = 0.125$$

$R_r = 12.5\%$

Step 4 - Calculate average code length

Symbol	Probability	Code	Length
S1	1/2	0	1
S2	1/4	10	2
S3	1/8	110	3
S4	1/8	111	3

Formula:

$$L = \sum p_i l_i$$

Substitution:

$$L = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$L = 1.75 \text{ bits}$$

Analysis & Conclusion:

Since: $L = H(X) = 1.75$ bits

The given code achieves the entropy lower bound & is optimal for the given source probabilities. It is also a prefix-free code because no codeword is the prefix of another codeword.

6. TASK 6 - Kullback-Leibler (KL) Divergence:

Given:

true distribution $P = (0.5, 0.25, 0.25)$

predicted distribution $Q = (1/3, 1/3, 1/3)$

Step1: calculate $KL(P||Q)$

Formula:

$$D_{KL}(P||Q) = \sum_i P(i) \log_2 \frac{P(i)}{Q(i)}$$

Substitution:

$$= 0.5 \log_2 \frac{0.5}{1/3} + 0.25 \log_2 \frac{0.25}{1/3} + 0.25 \log_2 \frac{0.25}{1/3}$$

$$= 0.5 \log_2 (1.5) + 0.5 \log_2 (0.75)$$

$$= 0.5 (0.5850) + 0.5 (-0.4150)$$

$$= 0.2925 - 0.2075$$

$$D_{KL}(P||Q) = 0.0850 \text{ bits}$$

Step2: calculate $KL(Q||P)$

Formula

$$D_{KL}(Q||P) = \sum_i Q(i) \log_2 \frac{Q(i)}{P(i)}$$

Substitution

$$= \frac{1}{3} \log_2 \frac{1/3}{0.5} + \frac{1}{3} \log_2 \frac{1/3}{0.25} + \frac{1}{3} \log \frac{1/3}{0.25}$$

$$= \frac{1}{3}\log_2\frac{2}{3} + \frac{2}{3}\log_2\frac{4}{3}$$

$$= \frac{1}{3}(-0.5850) + \frac{2}{3}(0.4150)$$

$$= -0.1950 + 0.2767$$

$$D_{KL}(Q||P) = 0.0817 \text{ bits}$$

Step 3: Analyse the asymmetry

The two KL divergence values are different:

Divergence	value
KL(P||Q)	0.0850 bits
KL(Q||P)	0.0817 bits

KL divergence is asymmetric because $D_{KL}(P||Q) \neq D_{KL}(Q||P)$

Conclusion:

KL divergence is a non-negative & asymmetric measure of difference b/w two probability distributions. Its direction must be selected carefully when it is used as a machine-learning training loss

---

7. TASK-7 - Entropy & Mutual Information

Given:

A	B	P(A,B)
0	0	0.4
0	1	0.1
1	0	0.1
1	1	0.4

1- Calculate H(A) & H(B)

Marginal probabilities:

$$P(A=0) = P(A=1) = 0.5$$

Similarly, $P(B=0) = P(B=1) = 0.5$

$$H(A) = H(B) = -2(0.5\log_2 0.5)$$

$$H(A) = H(B) = 1 \text{ bit}$$

2 Calculate Joint Entropy H(A,B)

$H(A,B) = -\sum P(A,B) \log_2 P(A,B)$

$= -2(0.4 \log_2 0.4) - 2(0.1 \log_2 0.1)$

$H(A,B) = 1.7219$ bits

3. Calculate conditional entropy $H(A|B)$

$H(A|B) = H(A,B) - H(B)$

$= 1.7219 - 1$

$H(A|B) = 0.7219$ bits

4. Calculate Mutual Information

$I(A;B) = H(A) + H(B) - H(A,B)$

$= 1 + 1 - 1.7219$

$I(A;B) = 0.2781$ bits

Interpretation:
A & B are dependent because their mutual information is greater than zero, knowing one variable provides information about the other

8. TASK 8 - Bayesian Decision Theory

Given:

$P(w_1) = 0.6$, $P(w_2) = 0.4$

$P(x|w_1) = N(2,4)$

$P(x|w_2) = N(6,4)$

Zero-one loss

1 Find the decision threshold

For equal variance the threshold is

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{P(w_1)}{P(w_2)}$$

$= 4 + \ln(1.5)$

$x^* = 4.4055$

2. Classify $x = 4.5$
Since $4.5 > 4.4055$, classify it as: $\boxed{\omega_2}$

3. Calculate probability of Error

$$P_e = 0.6\, P(x > 4.4055/\omega_1) + 0.4\, P(x \leq 4.4055/\omega_2)$$
$$= 0.6(0.1146) + 0.4(0.2126)$$
$$\boxed{P_e \approx 0.1538 = 15.38\%}$$

Conclusion:
Bayesian decision theory uses class probabilities & loss to make the best classification decision

9. TASK-9 Minimum Risk Decision

Given: $P(\omega_1/x) = 0.3$
$P(\omega_2/x) = 0.7$

where
ω_1 = disease
ω_2 = Healthy
α_1 = Declare disease
α_2 = Declare healthy

Loss matrix

Action	Disease (ω_1)	Healthy (ω_2)
Declare disease (α_1)	0	8
Declare healthy (α_2)	50	0

Step 1 – calculate conditional risk of α_1

Formula-
$$\boxed{R(\alpha_1/x) = \sum_j \lambda(\alpha_1/\omega_j)\, P(\omega_j/x)}$$

Substitution-
$$= 0(0.3) + 8(0.7)$$
$$\boxed{R(\alpha_1/x) = 5.6}$$

Step 2: Calculate conditional risk of α_2:

$$R(\alpha_2|x) = \sum_j \lambda(\alpha_2|\omega_j) P(\omega_j|x)$$

Substitution

$$= 50(0.3) + 0(0.7)$$

$$R(\alpha_2|x) = 15$$

Step 3: Determine minimum-risk action

Action	Conditional risk
declare disease	5.6
declare healthy	15

$5.6 < 15$

Choose α_1 (declare disease)

Step 4: Compare with MAP decision

The max A posteriori (MAP) decision selects the class having highest posterior probability

$P(\omega_1|x) = 0.3$, $P(\omega_2|x) = 0.7$

Since 0.7 is greater, MAP selects:

Declare healthy

However, minimum-risk decision selects declare disease

Step 5: Calculate posterior probability

$P = P(\omega_1|x)$

Risk of declaring disease: $R(\alpha_1|x) = 8(1-P)$

Risk of declaring healthy: $R(\alpha_2|x) = 50P$

for declaring disease to be optimal: $8(1-P) < 50P$

$8 - 8P < 50P$

$8 < 58P$

$P > 8/58$

$P > 0.1379$

∴ posterior probability threshold

$13.79\% \approx 13.8\%$

10 TASK-10 Bias-variance Trade-off & Generalisation

Given:

Polynomial degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1
Validation MSE	9.5	4.6	2.9	3.6	11.8

Step 1 - Calculate generalisation gap

Generalisation gap = validation MSE - Training MSE

for degree 1 - = 9.5 - 9.0 = 0.5

degree 2 = 4.6 - 4.1 = 0.5

degree 3 = 2.9 - 2.2 = 0.7

degree 5 = 3.6 - 0.9 = 2.7

degree 9 = 11.8 - 0.1 = 11.7

Generalisation gap table

Degree	Training MSE	Validation MSE	Generalisation gap
1	9.0	9.5	0.5
2	4.1	4.6	0.5
3	2.2	2.9	0.7
5	0.9	3.6	2.7
9	0.1	11.8	11.7

Step 2: Bias / variance analysis

degree	Analysis
1	high bias
2	Improved fit
3	Best balance b/w bias & variance
5	Beginning of overfitting
9	Severe overfitting & high variance

ect best model
st model is [polynomial degree 3]

ree 3 has lowest validation MSE of 2.9
its generalisation gap is relatively small.

Step 4 - Techniques to improve
1. Regularisation (L_1 or L_2)
2. Early stopping / model complexity reduction

Conclusion: A good model should balance bias & variance

11. TASK-11 Bayesian Estimation using Beta distribution

Given. 10 coins tosses with 7 heads & Beta (2, 2) prior

1. Maximum Likelihood Estimate (MLE):

$$MLE = \frac{7}{10} = 0.7$$

2. Maximum A posteriori (MAP):

Posterior distribution Beta (2+7, 2+3) = Beta (9, 5)

$$MAP = \frac{9-1}{9+5-2} = \boxed{0.667}$$

3. Posterior Mean

$$Mean = \frac{9}{9+5} = \frac{9}{14} = \boxed{0.643}$$

4. For 70 heads out of 100 tosses:

- MLE = 0.7
- MAP = 0.696
- Posterior Mean = 0.692

Conclusion: Bayesian estimation combines prior knowledge with observed data to estimate the probability of heads

Task-12 Sepsis prediction using Machine learning